

NUMA1.pdf



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Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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 **Gemini 3**
Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research x



 Fuentes v

 Archivos

Estudia con Gemini

SN 25/26

PARA LA COMUNIDAD
UNIVERSITARIA

EL SUBIDÓN

DE SIERRA NEVADA

Expert

2 DÍAS
CONSECUTIVOS

86€

Long expert

5 DÍAS
DE LIBRE USO

185€

Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

In a NUMA (Non-Uniform Memory Access time) multiprocessor there are two or more identical (NUMA) nodes, each one with a processor and its complete memory hierarchy, including a portion of main memory. The overall memory of the system is physically distributed among all the nodes but logically shared by all of them (i.e., the processor in any node can access to its main memory and also to any memory location in any other node through the interconnection network).

Which of the following statements are true?

A given physical memory address can only be stored in the memory of a single node, although multiple copies of the line containing that address may be temporarily stored in the cache memories of other nodes.

Trieu-ne una:

- ☒ Vertader
☐ Fals

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

In a NUMA system, instructions different from the conventional load and store are required to access to variables stored in other nodes.

Trieu-ne una:

- ☐ Vertader
☒ Fals

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

The way data is distributed among the different nodes of a NUMA multiprocessor system ...

Trieu-ne una:

- ☐ ... does not have any impact in the performance of the parallel application.
☒ ... is determined by the operating system based on a given data allocation policy (for example, first touch).
☐ ... dynamically changes with the objective of balancing the number of local accesses that are performed by the processors in the different NUMA nodes.
☐ ... is statically determined by the compiler, based on the accesses that are performed by the tasks in the parallel program.

Más info.:
elsubidon.es



WUOLAH

Pregunta 4

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

Assume that the coherence in a NUMA multiprocessor is based on a directory attached to the main memory in each node. The directory structure present in each node provides ...

Trieu-ne una:

- ☐ ... coherence information for the memory lines that are stored in the cache memory of the same node.
- ☐ ... information that allows a processor to find the data that is allocated in other nodes.
- ☒ ... information to keep coherent all possible copies in cache of the lines stored in the memory of that node.
- ☐ ... information that allows a processor in that node to find the nearest node where to find a given memory address, in order to minimize the memory access time.

Pregunta 5

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

In a NUMA multiprocessor system, with directory-based coherence protocol, the number of bits in each entry of the directory depends on the number of nodes in the system, with one or several additional bits to keep the state of the associated line.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals

Pregunta 6

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

The number of entries in the directory of a NUMA node ...

Trieu-ne una:

- ☐ ... is the total number of cache lines in the overall NUMA system, helping to identify which caches have a copy of a memory line.
- ☐ ... is determined by the maximum number of copies that are allowed for each line in main memory.
- ☒ ... is the number of lines that are stored in the main memory associated to it.
- ☐ ... depends on the number of NUMA nodes in the system in order to implement the list of nodes with remote copies.

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

the percentage of the whole main memory (**including both data and directory**) that is used by the directory to store all the information related to coherence?

Trieu-ne una:

- ☐ With the information provided one can not compute the number. You should have provided the size of the cache memory in each node to be able to compute the requested percentage.
- ☐ close to 200%
- ☒ close to 50%
- ☐ close to 100%

Pregunta 8

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

False sharing cannot occur across nodes in a NUMA multiprocessor architecture because the directory structure attached to main memory provides information about the location of each variable in the same line.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

Pregunta 9

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

In a NUMA multiprocessor architecture, false sharing implies the simultaneous existence of at least two copies of the same cache line in M state in the associated directory entry.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

Pregunta 10

Completa

Puntuació 1,00
sobre 1,00

[Marca la pregunta](#)

Two different processors in a cache-coherent multiprocessor architecture (either UMA or NUMA) continuously executing a count++ instruction originate a false sharing situation.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

WUOLAH